Solutions

1. What is vapour pressure?
1) The pressure exerted by the vapour of a liquid on its surroundings
2) The pressure exerted by a gas on its surroundings
3) The pressure exerted by a solid on its surroundings
4) The pressure exerted by a substance in its liquid or solid state on its surroundings

2. Which statement about vapour pressure is correct?
1) Vapor pressure decreases with increasing temperature.
2) Vapor pressure increases with increasing temperature.
3) Vapor pressure is independent of temperature.
4) Vapor pressure is directly proportional to the molecular weight of the substance.

3. Which substance would have the highest vapour pressure at a given temperature?
1) A substance with strong intermolecular forces
2) A substance with weak intermolecular forces
3) A substance with a high boiling point
4) A substance with a low boiling point

4. If vapour pressure of 12 gram of a liquid solution is "P" then what is the vapour pressure of 7 gram of same liquid
1) P
2) 2 P
3) P / 2
4) 4 P

5. If vapour pressure of 15 gram of a liquid solution is "P" then what is the vapour pressure of 9 gram of same liquid
1) P
2) 2 P
3) P / 2
4) 4 P

6. Liquid A has boiling point of 250 degree celcius and liquid B has boiling point of 120 degree celcius. What will be order of vapour pressure of liquid A and liquid B
1) $P_A = P_B$
2) $P_A > P_B$
3) $P_A < P_B$
4) None of these

7. Liquid X have the vapour pressure at 60 degree celcius is P_1 and 80 degree celcius is P_2 atm. Then what will be the order of vapour pressure of liquid X-
1) $P_1 = P_2$
2) $P_1 > P_2$
3) $P_1 < P_2$
4) None of these

8. H_2O and C_2H_5OH What will be the order of Vapour pressure
1) Vapour pressure of H_2O = Vapour pressure of C_2H_5OH
2) Vapour pressure of H_2O < Vapour pressure of C_2H_5OH

3) Vapour pressure of H_2O > Vapour pressure C_2H_5OH
4) Can't be compared

9. What will be the order of boiling point
1) HCl < HBr < HI < HF
2) HCl < HI < HBr < HF
3) HF < HCl < HBr < HI
4) HI < HBr < HC1 < HF

10. What will be the order of Vapour Pressure
1) HC1 < HBr < HI < HF
2) HF < HI < HBr < HCl
3) HF < HCl < HBr < HI
4) HI < HBr < HC1 < HF

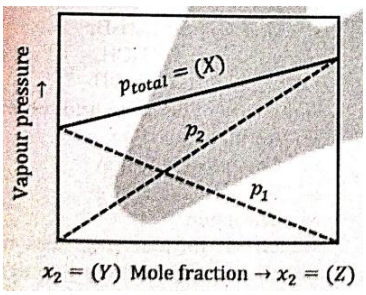
11. For a dilute solution, Roult's law states that:
1) the lowering of vapour pressure is equal to the mole fraction of solute
2) the relative lowering of vapour pressure is equal to the mole fraction solute
3) the relative lowering of vapour pressure is proportional to the amount of solute
4) the vapour pressure of the solution is equal to the mole fraction of solvent

12. Mixture of volume components A and B has total vapour pressure (in torr): $P = 254 - 119\,X_A$, where X_A IS MOLE FRACITON OF a IN THE mixture. Hence, P_A^0 and P_B^0 (in torr), respectively are
1) 254, 119
2) 135, 254
3) 119, 254
4) 154, 119

13. For a binary ideal liquid solution, the total pressure of the solution is given as:
1) $P_{total} = P_A^\circ + \left(P_A^\circ - P_B^\circ\right)X_B$.
2) $P_{total} = P_B^\circ + \left(P_A^\circ - P_B^\circ\right)X_A$
3) $P_{total} = P_B^\circ + \left(P_B^\circ - P_A^\circ\right)X_A$
4) $P_{total} = P_B^\circ + \left(P_B^\circ - P_A^\circ\right)X_B$

14. Two liquids A and B have vapour pressure in the ratio $P_A^o : P_B^o = 1:3$ at a certain temperature. Assume A and B form an ideal solution and the ratio of mole fractions of A to B in the vapour phase is 4 : 3, then the mole fraction of B in the solution at the same temperature is:
1) $\frac{1}{5}$
2) $\frac{2}{3}$
3) $\frac{4}{5}$
4) $\frac{1}{4}$

15. X, Y and Z in the above graph are



1) $X = p_1 + p_2, Y = 1, Z = 0$
2) $X = p_1 + p_2, Y = 0, Z = 1$
3) $X = p_1 \times p_2, Y = 0, Z = 1$
4) $X = p_1 - p_2, Y = 1, Z = 0$

16. If two substances A and B have $P_A^o : P_B^o = 1:2$ and have mole fraction in solution 1: 2 then mole fraction of A in vapours
1) 0.33
2) 0.25
3) 0.52
4) 0.2

17. In 1 mole heptane (V.P. = 92 mm of Hg) is mixed with 4 mol octane (V.P. = 31 mm of Hg), to form an ideal solution. Find out the vapour pressure of solution
1) 40
2) 35
3) 43.2
4) 54.8

18. For an ideal binary liquid solution with $P_A^o > P_B^o$, which relation between X_A (mole fraction of A in liquid phase) and Y_A (mole fraction of A in vapour phase) is correct?
1) $Y_A < Y_B$
2) $X_A > X_B$
3) $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$
4) $\frac{Y_A}{Y_B} < \frac{X_A}{X_B}$

19. An ideal solution is formed by mixing two volatile liquids A and B. X_A and X_B are the mole fractions of A and B respectively in the solution and Y_A and Y_B are mole fractions of A and в respectively in the vapour phase. A plot of $\frac{1}{Y_A}$ along y-axis against $\frac{1}{X_A}$ along x-axis gives a straight line. What is the slop of the straight line?
1) $\frac{P_B^0}{P_A^0}$
2) $\frac{P_A^0}{P_B^0}$
3) $P_B^0 - P_A^0$

4) $P_A^o - P_B^o$

20. A solution of liquids A and B having vapour pressure in pure state P_A^o and P_B^o. The solution contains 40% mole of A which is in equilibrium with 70% mole of A in vapour phase. If P_B^o is 260 torr then P_A^o is

1) 810 torr
2) 786 torr
3) 910 torr
4) 856 torr

21. A solution of acetone in ethanol

1) Behaves like a near-ideal solution
2) Obeys Raoult's law
3) Shows a negative deviation from Raoult's law
4) Shows a positive deviation from Raoult's law

22. A non-ideal solution was prepared by mixing 30 ml chloroform and 50 ml acetone. The volume of the mixture will be

1) > 80 ml
2) < 80 ml
3) = 80 ml
4) $\geq$ 80 ml

23. Which one of the following mixtures can be separated into pure components by fractional distillation

1) Benzene – toluene
2) Water - ethyl alcohol
3) Water - nitric acid
4) Water - hydrochloric acid

24. Identify the mixture that shows positive deviation from Raoult's law

1) $CHCl_3 + (CH_3)_2CO$
2) $(CH_3)_2CO + C_6H_5NH_2$
3) $CHCl_3 + C_6H_6$
4) $(CH_3)_2CO + CS_2$

25. All form ideal solutions except

1) C_2H_5Br and C_2H_5I
2) C_6H_5Cl and C_6H_5Br
3) C_6H_6 and $C_6H_5CH_3$
4) C_2H_5I and C_2H_5OH

26. A mixture of benzene and toluene forms

1) An ideal solution
2) Non-ideal solution
3) Suspension
4) Emulsion

27. Which of the following does not show negative deviation from Raoult's law

1) Acetone-Chloroform
2) Acetone-Benzene
3) Chloroform-Ether

4) Chloroform-Benzene

28. Liquids A and B form an ideal solution

1) The enthalpy of mixing is zero
2) The entropy of mixing is zero
3) The free energy of mixing is zero
4) The free energy, as well as the entropy of mixing, are each zero

29. One component of a solution follows Raoult's law over the entire range $0 \le x_1 \le 1$. The second component must follow Raoult's law in the range when x_2 is

1) Close to zero
2) Close to 1
3) $0 \le x_2 \le 0.5$
4) $0 \le x_2 \le 1$

30. Assertion: CCl_4 and H_2O are immisicible.

Reason: CCl_4 is a polar solvent.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
3) If assertion is true but reason is false.
4) If assertion is false but reason is true.

Answer Key

(1)	1	(16)	4
(2)	2	(17)	3
(3)	4	(18)	3
(4)	1	(19)	1
(5)	1	(20)	3
(6)	3	(21)	4
(7)	3	(22)	2
(8)	2	(23)	1
(9)	1	(24)	4
(10)	2	(25)	4
(11)	2	(26)	1
(12)	2	(27)	2
(13)	2	(28)	1
(14)	1	(29)	4
(15)	2	(30)	3